SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 3- CLASS TEST

Course Code:	**CSA1205**	**Course Title:**	**Computer Architecture for Multicore Systems**
CO Assessed:	**CO5**	**Assessment:**	**ASSESSMENT TOOL3- CLASS TEST**
Weightage:	**30%**	**Total Marks:**	**25**
CO5 Statement:	*Construct I/O communication mechanisms by comparing memory-mapped and I/O-mapped techniques, interrupt handling (vectored, hardware, software), DMA controllers, and advanced bus interfaces including PCIe, USB, NVMe, and Thunderbolt.*		
Student Name:	**SREYA.B**	**Reg.No.:**	**192521442**
Department:	**B.TECH IT**	**Date:**	

ANNEXURE A
CLASS TEST

1. Design a heterogeneous I/O bus architecture comparing USB, PCIe, and Thunderbolt, and construct a scalable communication model for multiple peripherals.

Heterogeneous I/O Bus Architecture

Design:

CPU → System Interconnect → PCIe / Thunderbolt / USB → Peripherals

Bus	Main Feature	Typical Use
USB	Low cost, plug-and-play	Keyboard, mouse, cameras
PCIe	Very high bandwidth, low latency	GPU, NVMe, network cards
Thunderbolt	High-speed, supports multiple protocols	Displays, storage, docking

Scalable model: Use PCIe for high-performance devices, Thunderbolt for external high-speed devices, and USB for general peripherals.

Result: Flexible communication with good scalability and efficient bandwidth allocation.

Construct an interrupt handling framework for a multi-core processor comparing vectored interrupts and shared interrupt lines, and analyze latency impact.

Multi-Core Interrupt Handling

Framework:

I/O Device → Interrupt Controller → CPU Core → Interrupt Service Routine (ISR)

Type	Description	Latency
Vectored Interrupt	Directly identifies the ISR address	Low
Shared Interrupt	Multiple devices share one interrupt line	Higher

Analysis: Vectored interrupts provide faster response because the processor directly identifies the required service routine. Shared interrupts require additional checking to identify the source.

Result: Use **vectored interrupts with per-core interrupt routing** for low-latency multi-core systems.

2. Design a high-speed NVMe storage subsystem using PCIe lanes, DMA transfers, and memory-mapped registers, and justify performance improvements.

High-Speed NVMe Storage Subsystem

Architecture:

CPU → PCIe → NVMe Controller → NAND Flash

PCIe lanes: Provide high-speed parallel data transfer.

DMA: Transfers data directly between NVMe and memory without continuous CPU involvement.

Memory-mapped registers: Allow the CPU to efficiently control the NVMe controller.

Multiple queues: Enable parallel I/O operations.

Result: Lower CPU overhead, reduced latency, high throughput, and faster storage access.

3. Construct an I/O scheduling mechanism comparing cycle stealing DMA and burst mode DMA for continuous multimedia streaming.

I/O Scheduling for Multimedia Streaming

Feature	Cycle Stealing DMA	Burst Mode DMA
Data transfer	Small portions	Large blocks
CPU interruption	More frequent	Less frequent
CPU availability	Higher	Lower during burst
Streaming performance	Moderate	High

Recommendation: Use **burst mode DMA** for continuous audio/video streaming because large blocks can be transferred efficiently with fewer interruptions.

Result: Higher throughput and smoother multimedia playback.

4. Design a multi-level I/O communication architecture combining memory-mapped I/O, interrupt handling, and advanced bus interfaces, and evaluate throughput.

Multi-Level I/O Communication Architecture

Architecture:

CPU → Cache/Memory → Memory-Mapped I/O → Interrupt Controller → PCIe/Thunderbolt/USB → I/O Devices

Memory-mapped I/O: Provides simple CPU access to device registers.

Interrupt handling: Notifies the CPU when an I/O operation requires attention.

PCIe: Used for high-speed devices.

Thunderbolt: Used for high-speed external devices.

USB: Used for general peripherals.

DMA: Transfers large amounts of data efficiently.

Evaluation: Combining these mechanisms reduces CPU overhead and improves overall I/O throughput, while high-speed buses such as PCIe handle bandwidth-intensive devices.

Code of Conduct:

I ___________YUVASRI.S (192521032)___________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Conceptual Understanding (30%)	**Accuracy of Answers (30%)**	**Problem Solving & Application Skills (20%)**	**Presentation & Organization (10%)**	**Completion & Time Management (10%)**
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	30%	Demonstrates excellent understanding of all concepts with accurate explanations and applications	Shows good understanding of concepts with minor errors	Shows basic understanding but with some misconceptions	Shows limited understanding with major conceptual errors
Accuracy of Answers	30%	Provides completely correct answers with proper steps, logic, and final solutions	Provides mostly correct answers with few minor mistakes	Provides partially correct answers with missing steps/errors	Provides incorrect or incomplete answers with minimal accuracy
Problem Solving & Application Skills	20%	Applies appropriate methods/techniques effectively and solves complex problems independently	Applies suitable methods with minor guidance/errors	Attempts problem solving but has difficulty applying concepts correctly	Unable to apply concepts or select suitable methods
Presentation & Organization	10%	Answers are well structured, neat, and clearly presented with proper notation/format	Answers are organized with minor presentation issues	Answers lack proper organization and clarity in some areas	Poorly organized answers with unclear presentation
Completion & Time Management	10%	Completes all questions within the allotted time with effective planning	Completes most questions within time	Attempts only part of the test due to time management issues	Very few questions attempted or incomplete submission